

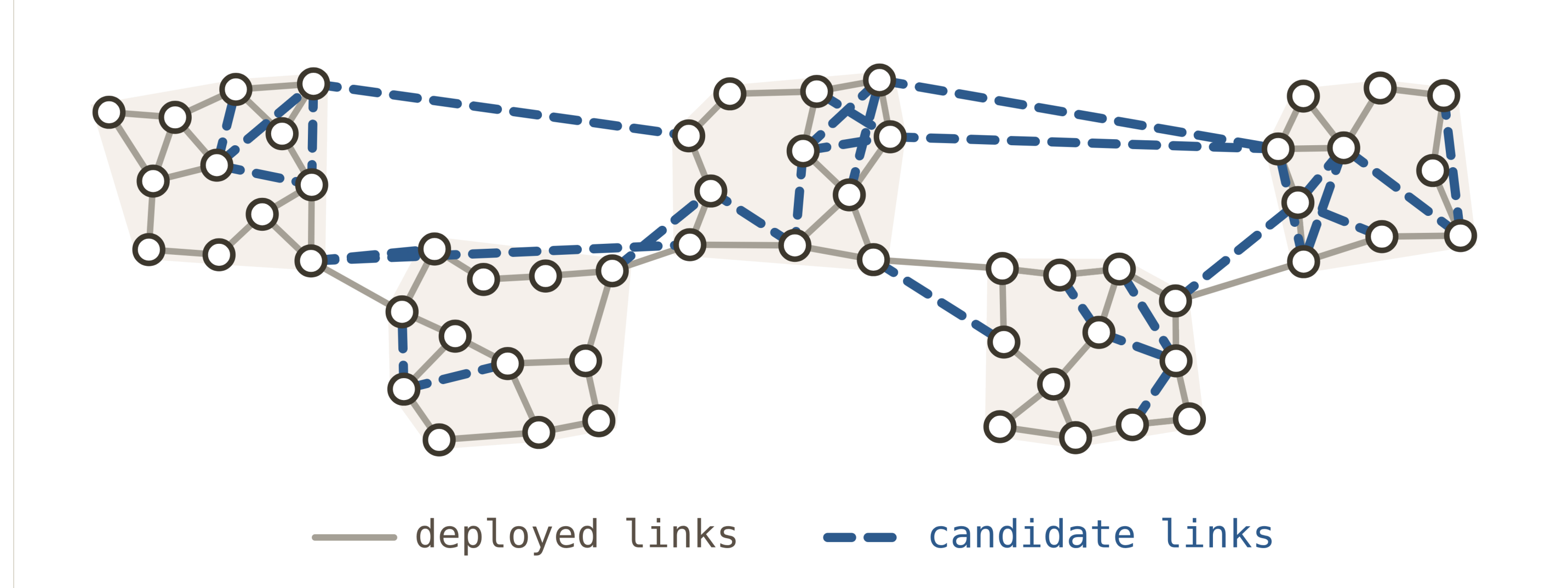
Cost-aware graph augmentation for robust networks via effective resistance

Omkar Bhoite[†] · V. Sateeshkrishna Dhuli[†] · Stefan Werner^{†,*} · Kimmo Kansanen[†]

[†] Department of Electronic Systems, Norwegian University of Science and Technology, Trondheim, Norway · ^{*} Department of Information and Communication Engineering, Aalto University, Espoo, Finland
{omkar.bhoite, venkata.s.dhuli, stefan.werner, kimmo.kansanen}@ntnu.no

01 Optimizing a pre-deployed network

Total effective resistance R_{tot} (the Kirchhoff index) is a tractable spectral proxy for robustness. Reducing it on a *deployed* weighted network means deciding **which links to add** and **how strong to make them** at once, one discrete choice coupled to a continuous one under a single global budget. Prices differ from candidate to candidate, combining a fixed per-link installation overhead with a penalty on the link's length.



The setup. Grey links and their weights are already deployed and **cannot be changed**. Dashed blue links form the candidate set $\mathcal{E}_c$. We select exactly k of them and size each within $[w_{\min}, w_{\max}]$. Every candidate carries its own unit price c_i and the total spend cannot exceed C .

OBJECTIVE

$$\mathbf{L}(\mathbf{w}) = \mathbf{L}_b + \sum_{e \in \mathcal{E}_c} w_e \mathbf{a}_e \mathbf{a}_e^\top$$

$$R_{\text{tot}}(\mathbf{w}) = n \operatorname{Tr} \left[\left(\mathbf{L}(\mathbf{w}) + \frac{\mathbf{1}\mathbf{1}^\top}{n} \right)^{-1} \right] - n$$

02 Joint selection and sizing

MIXED-INTEGER PROGRAM

$$\begin{aligned} & \underset{\mathbf{w}, \mathbf{z}}{\text{minimize}} && R_{\text{tot}}(\mathbf{w}) && \text{robustness of the upgraded network} \\ & \text{subject to} && \mathbf{c}^\top \mathbf{w} \leq C && \text{one global budget} \\ & && z_i w_{\min} \leq w_i \leq z_i w_{\max} && \text{perspective box, } z_i = 0 \Rightarrow w_i = 0 \\ & && \mathbf{1}^\top \mathbf{z} = k, \quad z_i \in \{0, 1\} && \text{exactly } k \text{ new links} \end{aligned}$$

(P₀)

03 A relaxation that certifies itself

Two obstacles make (P₀) intractable, the binary z_i and the matrix inverse inside R_{tot} . Relaxing $\mathbf{z}$ to $[0, 1]^p$ removes the first. Since $\mathbf{M}(\mathbf{w}) \succ 0$, bounding $\operatorname{Tr} \mathbf{M}^{-1}$ by $\operatorname{Tr} \mathbf{R}$ is *exactly* a Schur-complement condition. The inverse then disappears and the problem becomes convex.

SEMIDEFINITE SURROGATE

$$\begin{aligned} & \underset{\mathbf{R}, \mathbf{w}, \mathbf{z}}{\text{minimize}} && n \operatorname{Tr}(\mathbf{R}) - n \\ & \text{subject to} && \begin{bmatrix} \mathbf{M}(\mathbf{w}) & \mathbf{I} \\ \mathbf{I} & \mathbf{R} \end{bmatrix} \succeq 0 \\ & && \mathbf{c}^\top \mathbf{w} \leq C \\ & && z_i w_{\min} \leq w_i \leq z_i w_{\max}, \quad i = 1, \dots, p \\ & && \mathbf{1}^\top \mathbf{z} = k, \quad 0 \leq z_i \leq 1, \quad \mathbf{R} \in \mathbb{S}_+^n \end{aligned}$$

(P₁)

Rounding and repair. Keep the k largest entries of $\mathbf{z}^*$ and break ties toward cheaper links. Project the weights onto the induced box. If the design is over budget, scale toward $w_{\min} \hat{\mathbf{z}}$ until $\mathbf{c}^\top \hat{\mathbf{w}} \leq C$. The result is always buildable.

Proposition 1. Assume the base graph is connected, so $\mathbf{M}(\mathbf{w}) \succ 0$ for all feasible $\mathbf{w}$. Then (P₁) satisfies $R_{\text{tot}}^{\text{SDP}} \leq R_{\text{tot}}^{\text{MI}}$. Moreover, at any optimum $(\mathbf{R}^*, \mathbf{w}^*, \mathbf{z}^*)$ of (P₁), $\mathbf{R}^* = \mathbf{M}^{-1}(\mathbf{w}^*)$. Consequently, (P₁) provides a valid lower bound, and paired with any feasible discrete augmentation it yields an instance-wise optimality gap.

Remark 1. Given a feasible rounded pair $(\hat{\mathbf{w}}, \hat{\mathbf{z}})$, we have

$$R_{\text{tot}}^{\text{SDP}} = \text{val}(\mathcal{P}_1) \leq \text{val}(\mathcal{P}_0) = R_{\text{tot}}^{\text{MI}} \leq R_{\text{tot}}(\hat{\mathbf{w}}) =: R_{\text{tot}}^{\text{RND}}$$

so the relative gap ratio $(R_{\text{tot}}^{\text{RND}} - R_{\text{tot}}^{\text{SDP}}) / R_{\text{tot}}^{\text{SDP}}$ measures solution quality.

04 Solved as a cone program

Scaled vectorization stacks the unknowns as $\mathbf{x} = [\text{svec}(\mathbf{R})^\top \mathbf{w}^\top \mathbf{z}^\top]^\top$. Every constraint of (P₁) then asks only that $\mathbf{x}$ lie in one product cone.

$$\mathcal{K} = \underbrace{\mathbb{S}_+^{2n}}_{\text{LMI}} \times \underbrace{\mathbb{R}_+}_{\text{budget}} \times \underbrace{\mathbb{R}_+^p \times \mathbb{R}_+^p \times \mathbb{R}_+^p \times \mathbb{R}_+^p}_{\text{perspective box and } z \text{ bounds}} \times \underbrace{\{0\}}_{\text{cardinality}}$$

- Solved by ADMM on the homogeneous self-dual embedding (SCS).
- A cached $\mathbf{LDL}^\top$ factorization reduces each affine step to two triangular solves.

05 Greedy baseline with rank-one updates

The Sherman and Morrison rank-one update identity gives each marginal gain in closed form. The inverse $\mathbf{M}^{-1}$ is formed once and then updated after every insertion rather than recomputed. Each trial weight is priced by a **tentative budget allocation** $\mathbf{s} = C_{\text{rem}}/k_{\text{rem}}$.

MARGINAL GAIN, EXACT

$$\Delta R_{\text{tot}} = \frac{n w_{(v_i, v_j)} b^2(v_i, v_j)}{1 + w_{(v_i, v_j)} r(v_i, v_j)}$$

$$\mathbf{r} = \mathbf{a}^\top \mathbf{M}^{-1} \mathbf{a}$$

TRIAL WEIGHT

$$\mathbf{w}_e = \max\{w_{\min}, \min\{w_{\max}, s/c_e\}\}$$

from the tentative allocation $\mathbf{s} = C_{\text{rem}}/k_{\text{rem}}$

$$b^2 = \|\mathbf{M}^{-1} \mathbf{a}\|_2^2 \quad \text{total cost } \mathcal{O}(n^3 + k(pn + n^2))$$

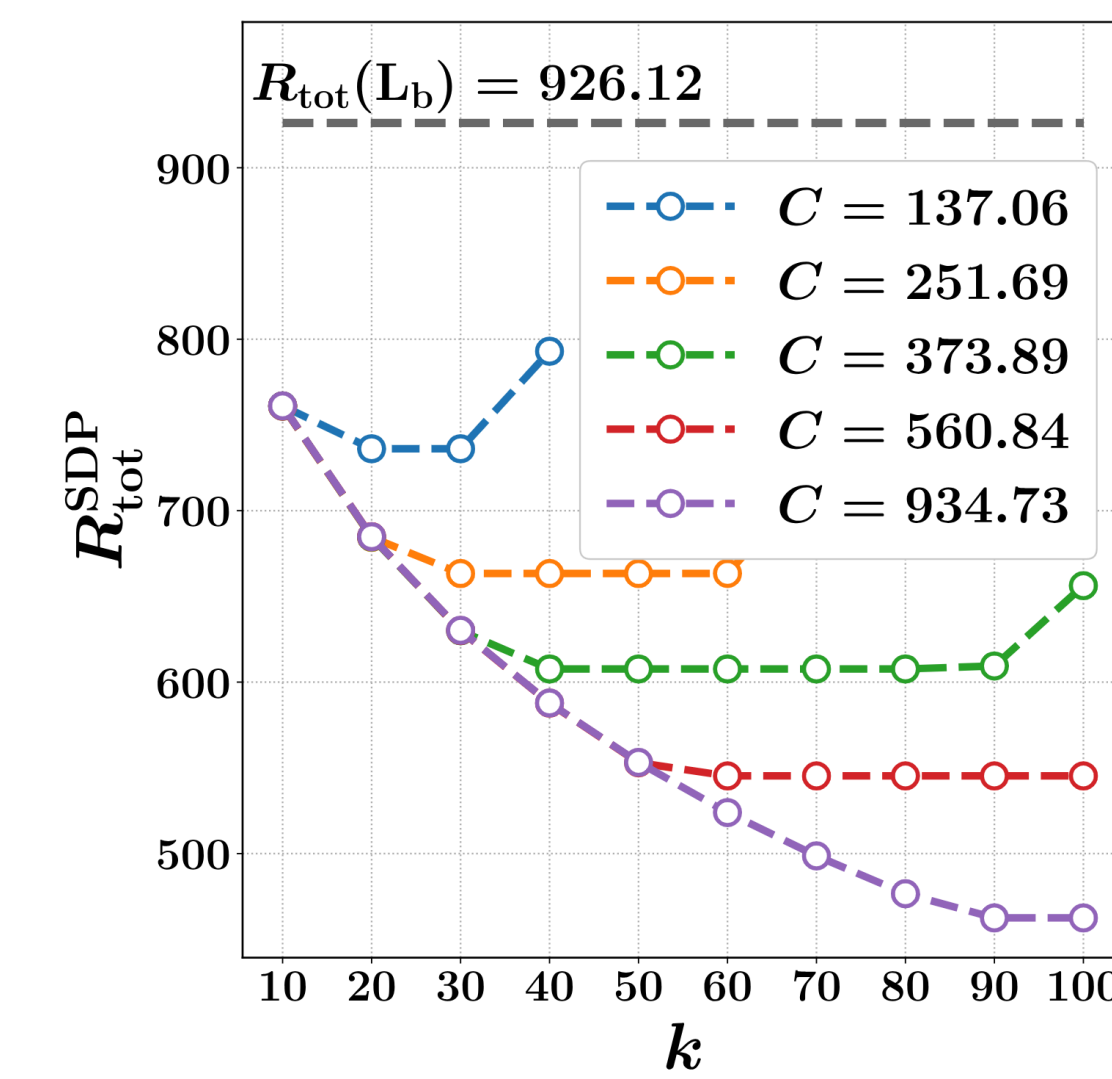
06 Instances and candidate sets

Synthetic. Base graph of $n = 100$ planar vertices with lognormal weights. Candidate weights span the 0.75 to 0.95 quantiles. Three k-means regions give $k_{\text{in}} = 3$ neighbour links and a sparse backbone.

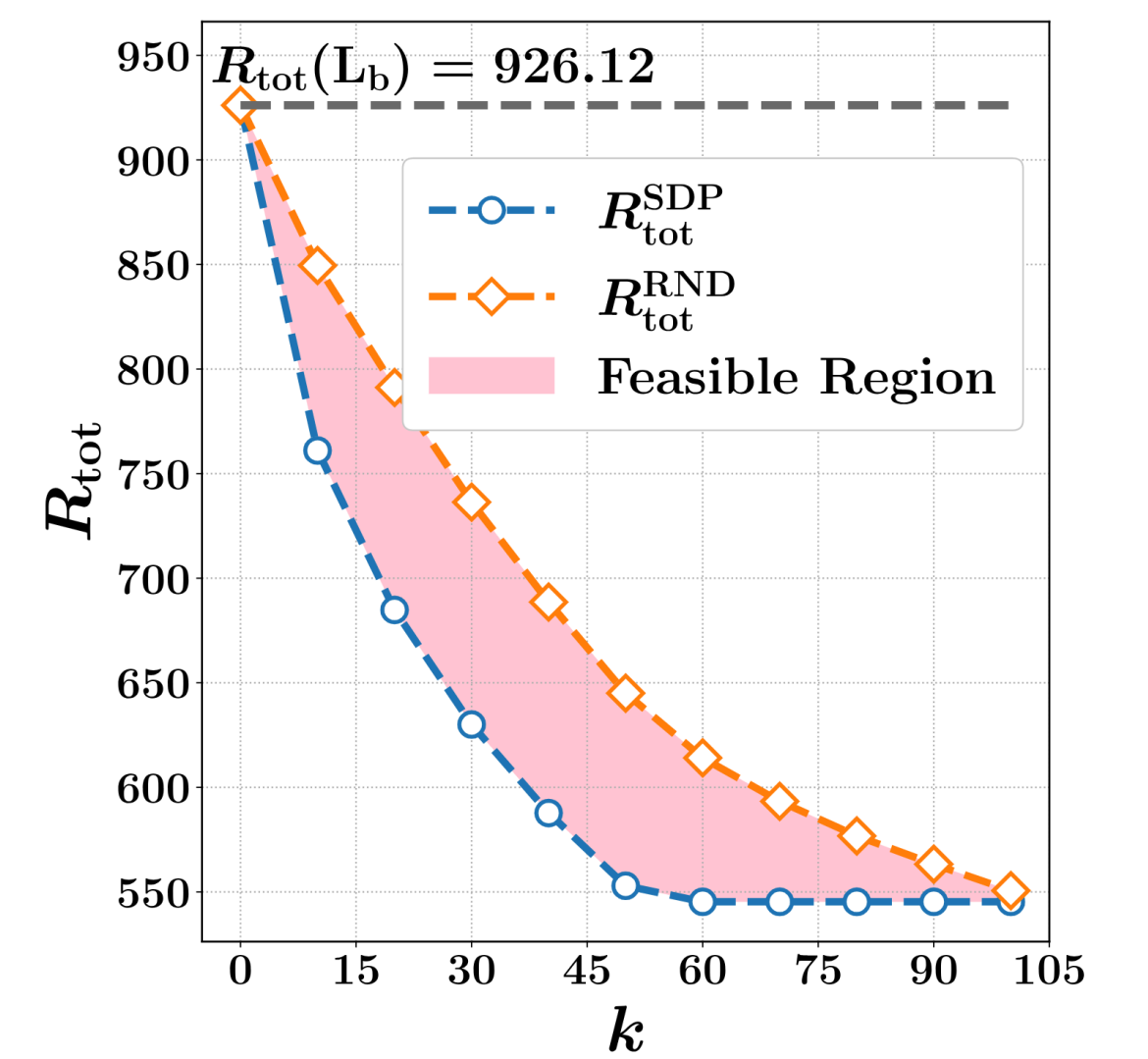
Cost model. A fixed-charge model, $\mathbf{c} = c_{\text{fixed}} \mathbf{1} + \beta \mathbf{d}_c$. Here c_{fixed} is the per-link installation overhead. Lengths in $\mathbf{d}_c$ are normalised by their mean.

Budget regimes. Budgets are deduced from the candidate cost structure. The least budget admitting k links is $C_{\min}(k) = w_{\min} \sum_{i=1}^k c_{(i)}$, summed over the cheapest candidates. Five levels centre on the anchor $C_{\min}(k_{\max}) = 373.89$, two at the thresholds $k_{\max}/3$ and $2k_{\max}/3$ and two at $1.5\times$ and $2.5\times$.

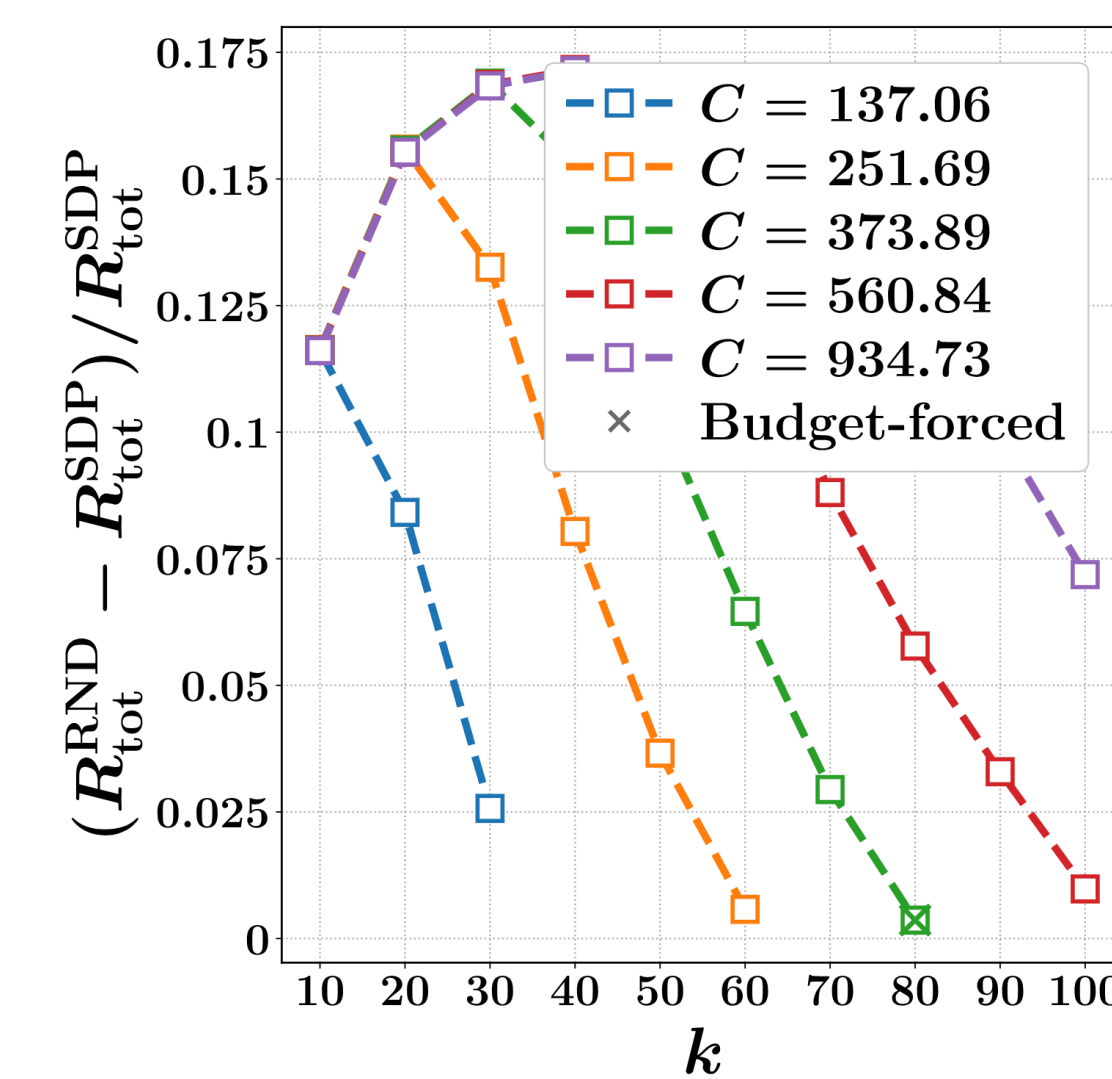
07 What the budget buys



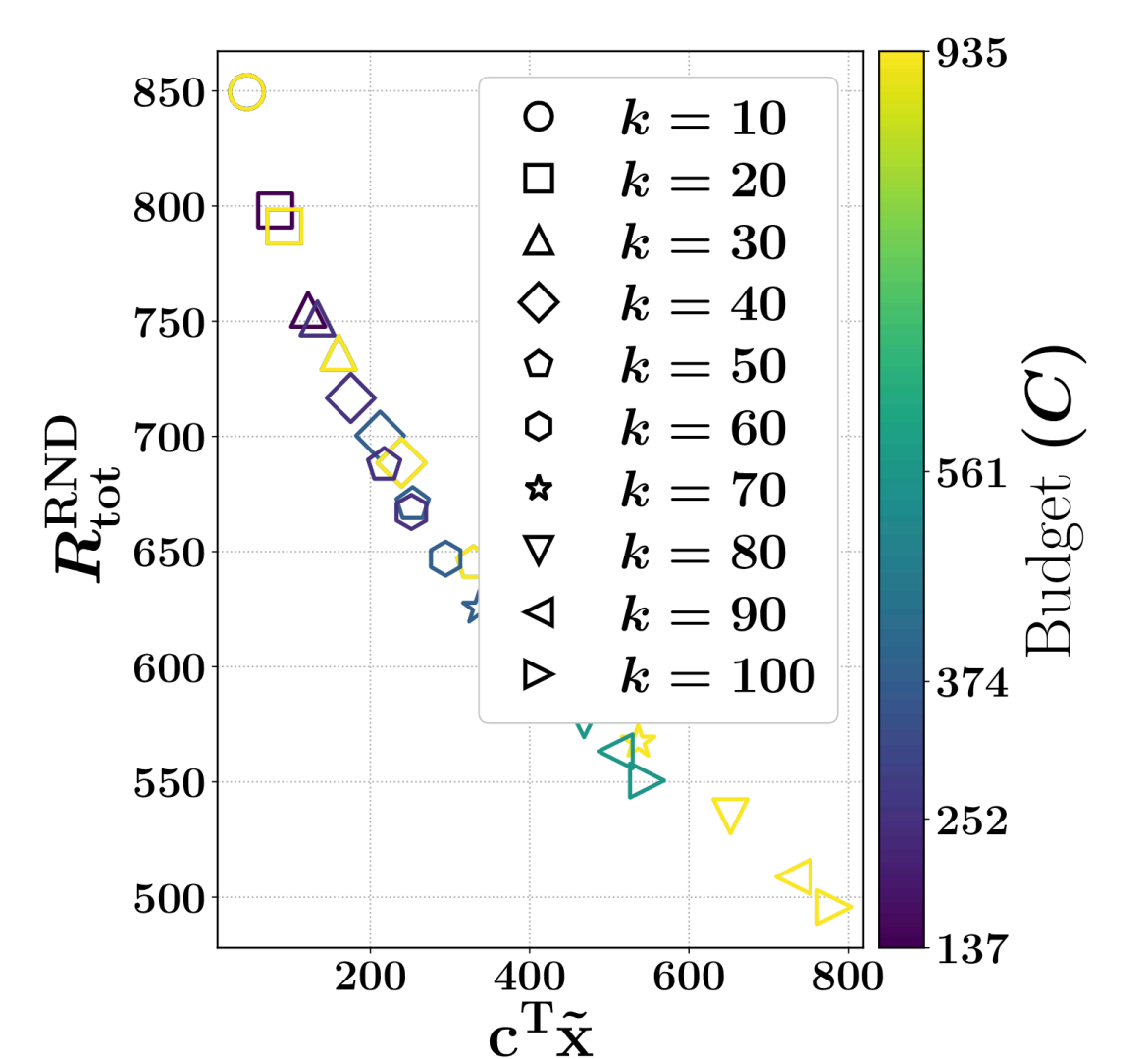
1a. $R_{\text{tot}}^{\text{SDP}}$ versus k at different budgets.



1b. Feasible region between the bound and the rounded design.

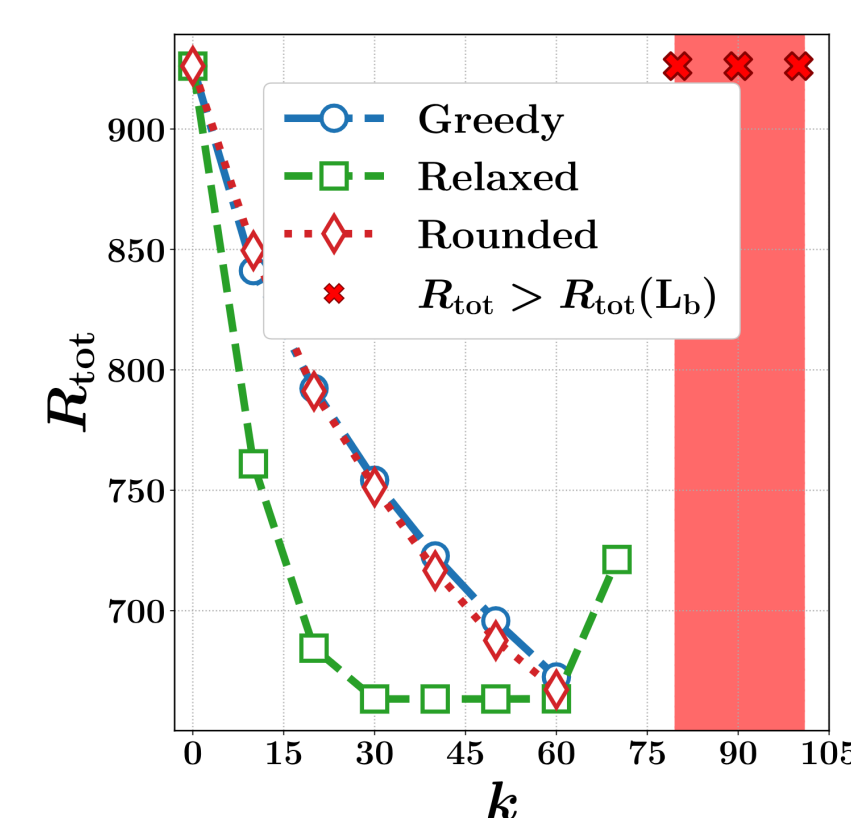


1c. Relative gap $(R_{\text{tot}}^{\text{RND}} - R_{\text{tot}}^{\text{SDP}}) / R_{\text{tot}}^{\text{SDP}}$ at each budget.

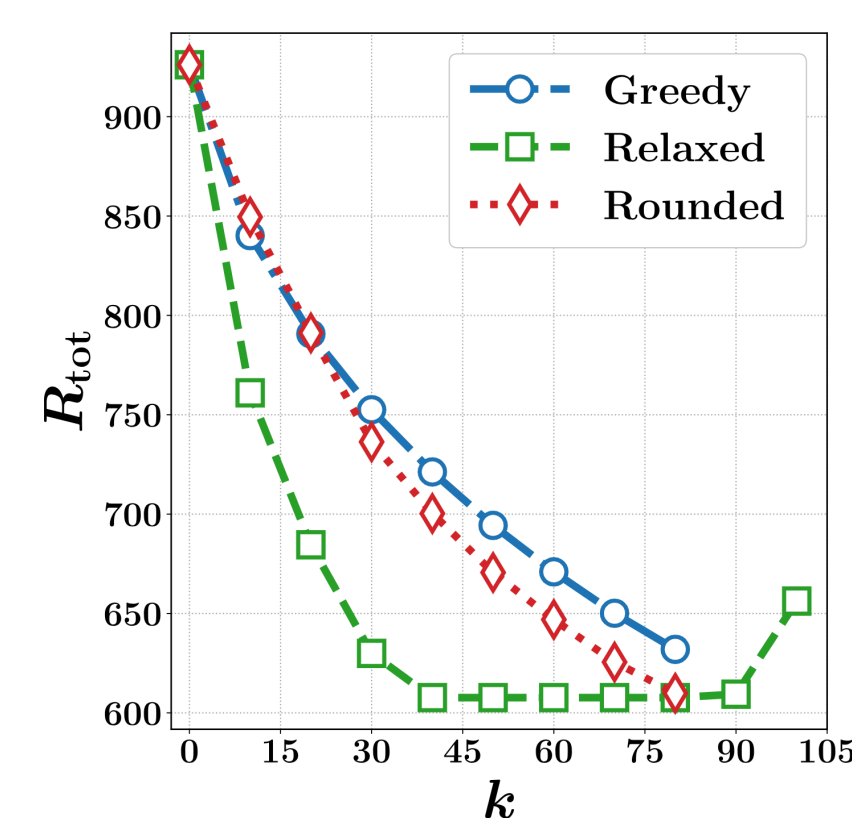


1d. Pareto front against budget spent at fixed k .

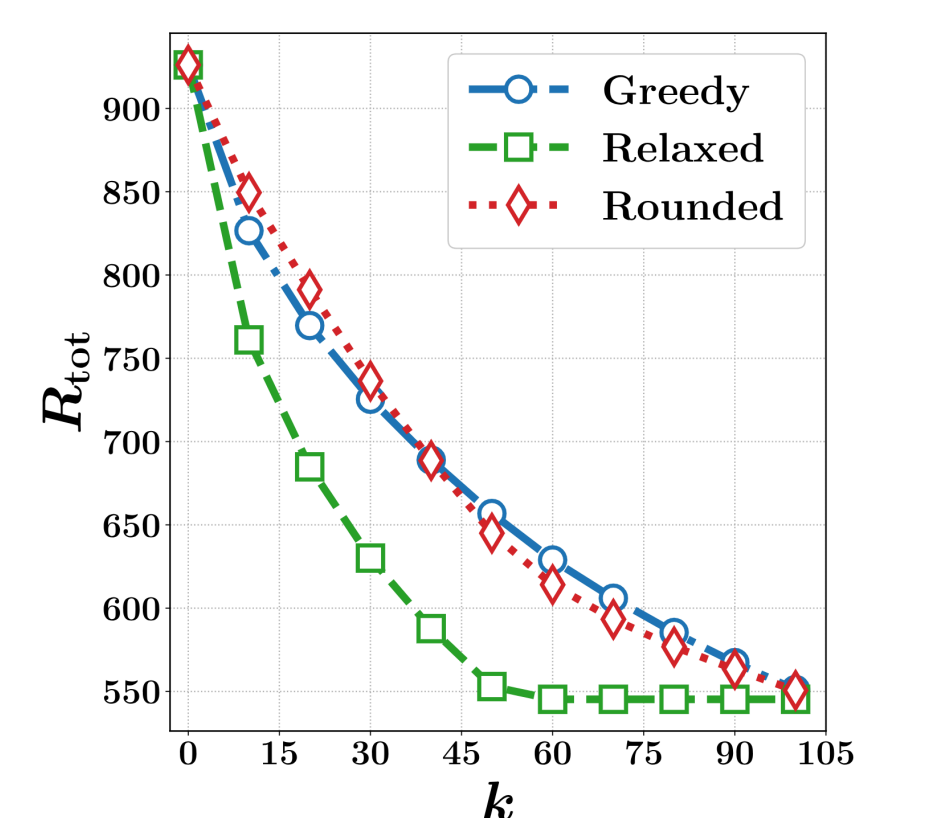
08 Budget regimes



Constrained. $C = 251.69$. Infeasible from $k = 80$. Greedy stalls after 61 links.

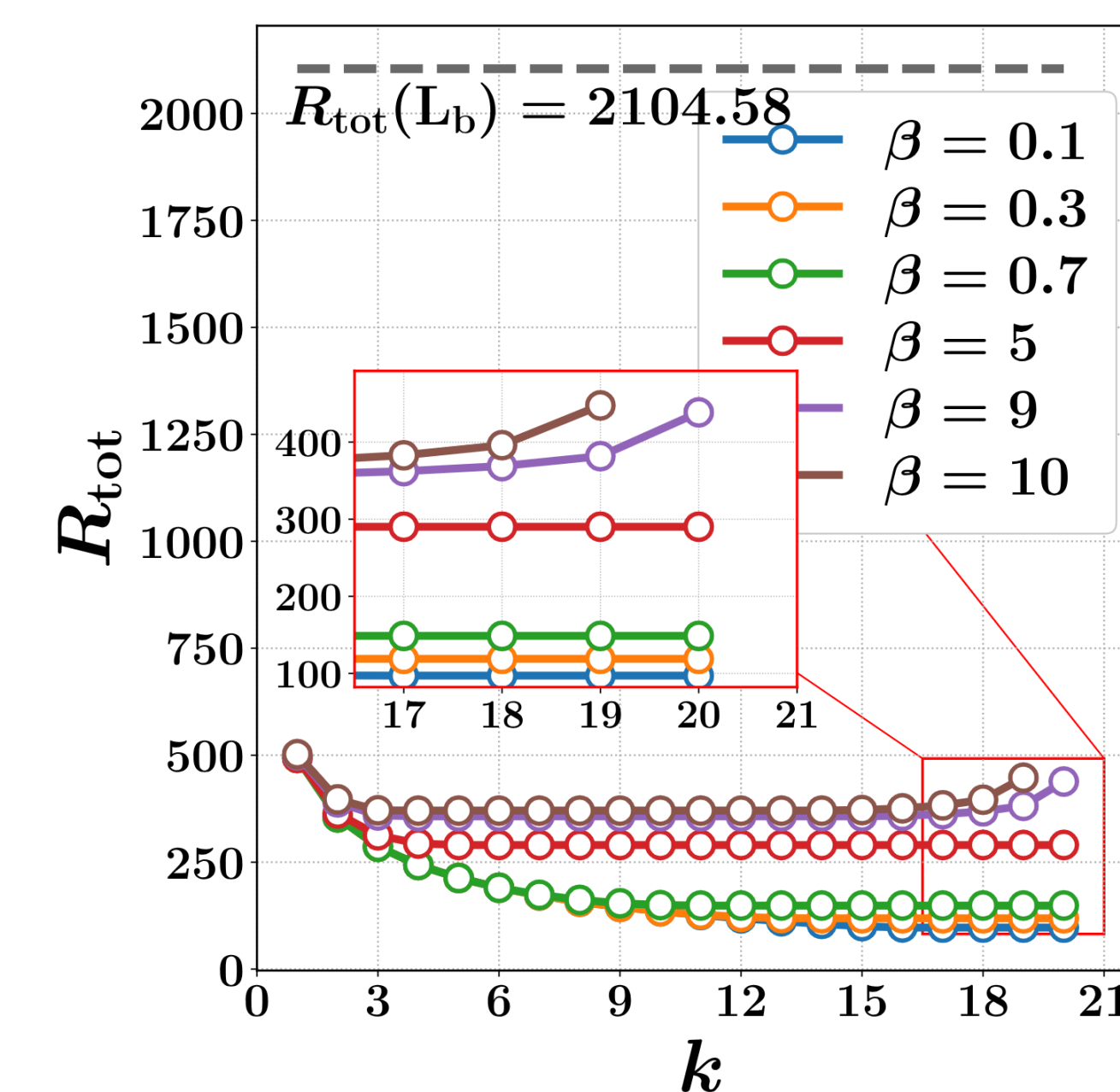


Marginal. $C = 373.89$, the anchor. Feasible but on the boundary.



Relaxed. $C = 560.84$. All k feasible. Rounded tracks the bound.

09 California road network



2a. $n = 500$ sampled nodes at $C = 150$, sweeping β . A steeper β lifts the curve.

10 Takeaways

- The semidefinite relaxation lower-bounds every feasible discrete augmentation.
- The Pareto front exposes the trade-off between effective resistance and budget spend.
- The relative gap between relaxed and rounded designs measures the loss from discretization.
- The rank-one greedy trades solution quality for computational speed.

